

AYMANE AIT DADS

AI & Wireless Research Intern | LLM Fine-Tuning · Intelligent Systems Prototyping · Signal-Aware Deep Learning

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PROFESSIONAL SUMMARY

Fine-tuned a 30B Mamba-Transformer MoE model (Nemotron) with LoRA adapters achieving top 10% on the Kaggle leaderboard (0.80/1.0, \$106K+ prize pool), demonstrating deep expertise in large-model training and ablation-driven research methodology directly applicable to 6G AI prototyping. Ranked 1st in EURECOM class for AI-generated image detection (0.791 AUC) using CLIP ViT-L/14, with submission to the NTIRE 2026 @ CVPR international benchmark, evidencing academic-benchmark-level engineering rigor. Core competencies span LLM fine-tuning with PEFT/LoRA, transformer architectures, Python, PyTorch, and Hugging Face Transformers - all cited in the target role requirements. Available for a 6-month internship from summer 2025, based in Sophia Antipolis, France, with full authorization to work in Germany and the European Union.

CORE COMPETENCIES

AI Prototyping | Intelligent Systems | Transformer Architecture | LLM Fine-Tuning | Ablation Studies | Signal Processing | Next-Generation Wireless | Research Engineering

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a Random Forest + K-Means pipeline on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes, directly informing infrastructure optimization.
- Designed a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with model output adopted directly by the network optimization team for production decision-making.
- Reduced manual network analysis time by **~60%** through automated feature engineering, model evaluation pipelines, and reproducible Python-based reporting workflows built with Pandas and Scikit-learn.
- Delivered stakeholder presentations translating **Random Forest model outputs** into actionable maintenance recommendations, bridging technical findings and business strategy for non-technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - LLM Fine-Tuning *Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard (0.80/1.0)*

- Fine-tuned a **30B Mamba-Transformer MoE model** (3.5B active params via MoE routing) with LoRA ($r=32$, ~888M trainable parameters) achieving 0.80/1.0 on the public leaderboard (top 10%, \$106K+ prize pool), using SFTTrainer with completion-only loss on 7,828 verified chain-of-thought reasoning examples across 6 logic puzzle types.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patching CUDA allocators, stubbing Mamba/Cutlass imports, disabling fast-path CUDA kernels, fixing Triton ptxas permissions, and replacing rmsnorm with a PyTorch fallback - while conducting 20+ single-variable ablations confirming that packing=True reduced score from 0.80 to 0.64 and synthetic data consistently degraded performance.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - CLIP ViT Fine-Tuning *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class (0.791 AUC)*

- Fine-tuned **CLIP ViT-L/14 (428M params)** with a custom MLP head (768-512-256-1, GELU, BatchNorm, Dropout) on 250K images spanning 25+ generator types, achieving 0.791 AUC and ranking 1st on the private EURECOM class Kaggle leaderboard, with a separate submission to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Identified through ablation that **1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC)** due to out-of-distribution generalization, and boosted ensemble performance via 10-view TTA (flips, crops, blur, resizes) weighted by validation AUC using dual learning rates (2e-6 CLIP backbone, 5e-4 head) with FP16 mixed precision.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision · Kaggle GPU

Anomalous Sound Detection - Industrial Equipment

EURECOM · 2025 · Unsupervised Fault Detection

- Designed an unsupervised **Transformer autoencoder on Mel spectrograms** with SpecAugment for predictive maintenance of slide-rail industrial machinery, achieving AUC 0.80 with no labeled anomaly data available during training.
- Applied signal-processing-aware data augmentation and reconstruction-error thresholding to detect **out-of-distribution acoustic signatures**, producing a fully unsupervised anomaly detection pipeline applicable to sensor-rich intelligent systems.

PyTorch · Mel Spectrograms · SpecAugment · Transformer Autoencoder

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Research: LoRA · QLoRA · SFTTrainer · PEFT · Mamba-Transformer · MoE · vLLM · chain-of-thought · ablation studies

Deep Learning & Computer Vision: PyTorch · Hugging Face Transformers · OpenCLIP · CLIP ViT · CNN · DenseNet · FP16/BF16 mixed precision · Triton

Data Science & MLOps: Python · Scikit-learn · Pandas · NumPy · feature engineering · Git · Linux · CUDA debugging · Power BI